



Research Paper

Isolated Intracranial Hypertension as a Presentation of Pediatric Lyme Borreliosis: A Case Report and Literature Review

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ABSTRACT

Background: It is extremely rare for Lyme borreliosis to present solely with features of increased intracranial pressure. The treatment of pediatric Lyme neuroborreliosis with oral versus intravenous antibiotics remains controversial.

Methods: Case report and literature review.

Results: A 13-year-old male presented with five days of binocular diplopia, several weeks of headache, and a history of multiple tick bites six weeks prior. His examination showed a left eye abduction deficit and bilateral optic disc edema. Magnetic resonance imaging (MRI) of the brain with contrast showed tortuosity of the optic nerves, prominence of the optic nerve sheaths, and enhancement of the left fifth and bilateral sixth cranial nerves. Lumbar puncture showed an elevated opening pressure and a lymphocytic pleocytosis. Lyme IgM and IgG antibodies were positive in the serum and cerebrospinal fluid. The patient was treated with intravenous ceftriaxone for two days empirically followed by doxycycline by mouth for 19 days. Symptoms began improving after 48 hours. The strabismus resolved after two weeks, and the papilledema improved slowly with complete resolution at six months.

Conclusions: Lyme neuroborreliosis can present as isolated intracranial hypertension in the pediatric population; it can be differentiated from idiopathic intracranial hypertension on MRI, and lumbar puncture and can be confirmed with serum antibody testing. Oral doxycycline can be considered for Lyme neuroborreliosis in children.

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Introduction

Lyme borreliosis is a relatively common infectious disease caused by the spirochete *Borrelia burgdorferi* and transmitted by ticks of the *Ixodes ricinus* complex.^{1,2} Without antibiotic therapy, the disease occurs in three stages. Stage 1 is early, localized disease consisting of a skin infection at the site of the tick bite; it typically appears as a circular, expanding lesion termed *erythema migrans*. Stage 2 is early, disseminated infection where *B. burgdorferi* spreads to other organs. Infection of the nervous system in this stage results in Lyme neuroborreliosis. Stage 3 is late, localized infection most commonly manifesting as Lyme arthritis.³ Lyme borreliosis

presents as erythema migrans in 80%, followed by nonspecific symptoms without a rash in 18%, and early or late disseminated infection in the remaining 2% to 3%.⁴ Herein, we describe the case of a pediatric patient presenting with isolated intracranial hypertension (IIH) secondary to Lyme neuroborreliosis.

Materials and Methods

We performed a case report and literature review. We searched Medline and EMBASE on May 29, 2023, from their inception using the following medical subject headings: (Lyme disease OR *Borrelia burgdorferi* OR *Borrelia* infection) AND (intracranial hypertension OR brain pseudotumor).

Results

A healthy 13-year-old male presented with five days of horizontal binocular diplopia and several weeks of headache. His initial examination showed a mild abduction limitation of the left eye and

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an esotropia that worsened on left gaze consistent with a left cranial nerve VI palsy. His dilated fundus examination showed bilateral optic disc edema (Fig). Optical coherence tomography of the optic discs reaffirmed the finding of edema (Supplemental Figure 1). The remainder of his ophthalmic and cranial nerve examination was unremarkable.

Magnetic resonance imaging (MRI) of the brain with contrast showed mild tortuosity of the optic nerves and prominence of the optic nerve sheaths bilaterally and enhancement of the left fifth and bilateral sixth cranial nerves. Further history revealed multiple tick bites while in the St. Lawrence River Valley of Ontario, Canada, six weeks prior. Lumbar puncture performed in the left lateral decubitus position showed an opening pressure of 49.3 cm H₂O and a lymphocytic pleocytosis ($36 \times 10^6/L$, 84% lymphocytes). Cerebrospinal fluid (CSF) glucose (3.1 mmol/L) and protein (0.51 g/L) were normal. Lyme IgM and IgG serum and CSF antibodies were positive (enzyme-linked immunosorbent assay and Western blot).

The patient was treated with intravenous ceftriaxone 2 g daily for two days empirically and then transitioned to doxycycline 100 mg by mouth twice daily for 19 days for treatment of Lyme neuroborreliosis. Symptoms began improving after 48 hours. The strabismus resolved after two weeks. The papilledema improved steadily and resolved by six months.

Discussion

Lyme borreliosis presenting as IIH in the pediatric population is extremely rare. In our review of the literature, we identified 15 reports describing 25 total cases of Lyme neuroborreliosis presenting with IIH.^{5–19} When it occurs, it can be difficult to differentiate from idiopathic IIH on clinical examination alone. Of the 25 cases identified in our literature review, 16 had additional clinical features suggesting an alternative diagnosis to IIH: 11 had skin rash, two had arthritis symptoms, three had meningitis symptoms, and four had focal neurological signs not attributable to increased intracranial pressure (ICP) (Supplemental Table 1). In the remaining nine cases the features helpful for distinguishing Lyme neuroborreliosis from IIH were MRI with abnormal findings not attributable to increased ICP in five cases and lumbar puncture with abnormal CSF composition in seven cases (Supplemental Table 2). In all cases except one, antibody testing was positive (Supplemental Table 3).

Our case and literature review highlight the importance of performing and carefully reviewing the results of a contrast-enhanced MRI of the brain and a lumbar puncture in pediatric patients with papilledema to exclude secondary causes of elevated ICP including Lyme neuroborreliosis. White matter lesions and nerve root or meningeal enhancement on MRI of the brain with contrast have been reported in Lyme neuroborreliosis and can distinguish these cases from IIH.²¹ Lumbar puncture also serves an important role in cases of Lyme neuroborreliosis with inflammatory CSF changes including pleocytosis, elevated protein, and intrathecal immunoglobulin synthesis expected with few exceptions.²²

When Lyme neuroborreliosis is suspected, two-tier serum antibody testing performed with enzyme-linked immunosorbent assay and confirmed with Western blot is recommended.²³ Occasionally, patients are seronegative at presentation; this typically occurs when the disease duration is less than six to eight weeks.^{24,25} In these instances, repeat testing after two to four weeks is recommended as the sensitivity increases to 98% with disease duration greater than eight weeks.^{24–31} Obtaining simultaneous samples of serum and CSF at presentation to determine the CSF: serum antibody index is also recommended.³² An elevated CSF:serum index is more specific than serum antibody testing for Lyme neuroborreliosis and in rare instances, may be positive before serum antibody testing.^{23,24,33–38} The utility of the CSF:serum index is limited in patients with previous Lyme neuroborreliosis as it can stay elevated for years after treatment.^{39–41} In these cases, other measures of CSF inflammation, such as CSF leukocyte count, can aid in distinguishing between active and prior infection.⁴² Culture or polymerase chain reaction testing on serum or CSF is not recommended.⁴² The measurement of the chemokine CXCL13 in the CSF is of uncertain diagnostic value at this time and is not routinely recommended.⁴²

The treatment of pediatric Lyme neuroborreliosis remains controversial. Traditionally, intravenous antibiotics have been favored, owing to the higher quality of evidence existing for parenteral regimens when guidelines were first published.⁴³ However, recently the literature supporting oral doxycycline has grown.^{44–48} The association between doxycycline and tooth staining has also been questioned.²³ Updated treatment guidelines now state that oral doxycycline may be considered over parenteral therapy for Lyme neuroborreliosis without parenchymal involvement in all children who can tolerate oral antibiotics.⁴²

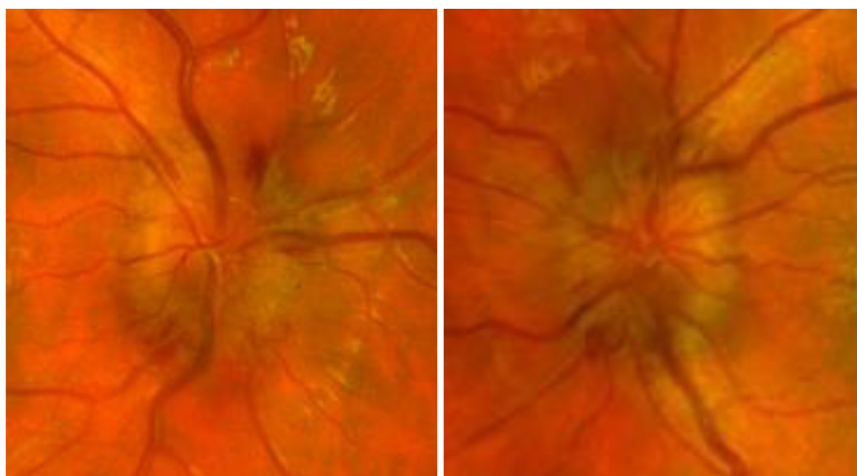


FIGURE. Pseudocolor photographs of the optic discs at presentation. Features of optic disc edema including 360° elevation and blurring of the optic disc margins, obscuration of minor vessels, and disc hemorrhages are demonstrated bilaterally. The color version of this figure is available in the online edition.

Acetazolamide can be considered as an adjunct to antibiotics in Lyme neuroborreliosis for the treatment of secondary IIH; it was not thought to be necessary in our case as the patient had no vision loss and the clinical signs of increased ICP were improving with antibiotics alone.

Conclusions

Lyme neuroborreliosis can present with IIH in the absence of a tick bite history and clinical manifestations of Lyme disease. In these cases, contrast-enhanced brain MRI and CSF analysis can exclude IIH and provide valuable clues to prompt further testing for Lyme neuroborreliosis. When Lyme neuroborreliosis is suspected, two-tier serum antibody testing and determination of the CSF: serum antibody index are recommended to confirm the diagnosis. If initial testing is negative but suspicion remains, repeat serum antibody testing is recommended after two to four weeks. Oral doxycycline can be considered as a treatment for Lyme neuroborreliosis.

CRediT authorship contribution statement

Jeffrey M. Mah: Conceptualization, Data curation, Formal analysis, Writing – original draft. **Cody Lo:** Data curation, Writing – original draft. **Michael D. O'Connor:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary Data

Reference 20 is cited in the Supplemental Table 3 footnote.

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.pediatrneurol.2023.12.025>.

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